Practice 1 Report: Analysis of Evaluation Metrics in Machine Learning

Classification Problems

In this section I analysed four binary classification models (A, B, C, and D). The models were evaluated using a threshold of 0.5 to convert predicted probabilities into binary classes. And the analysis was made using no external libraries (only Python, Pandas, and Numpy).

Calculated Metrics

Metric	Model A	Model B	Model C	Model D
Accuracy	0.5000	0.9153	0.2803	0.7674
Recall	0.7633	0.2662	0.2168	0.8806
Precision	0.5000	0.7008	0.2484	0.8372
TNR	0.2367	0.9874	0.3438	0.3148
F1-Score	0.6042	0.3859	0.2315	0.8583
AUC-ROC	0.4996	0.6300	0.2753	0.6008

Interpretation and Analysis

- **Model A:** Achieving an Accuracy of 0.50 and a AUC-ROC of 0.4996, the model performs nothing better than random probability. Despite getting a good Recall (0.76), it has a low TNR (0.23) which indicates it is predicting the positive class too frequently, thus, leading to a high rate of False Positives.
- **Model B:** This model shows a high Accuracy (0.91) but a quite low F1-Score (0.38). Due to class imbalance; the model is good at identifying the negative class (TNR 0.98) but performs poorly on identifying the positive class (Recall 0.26).
- **Model C:** Its AUC-ROC of 0.27 shows this model is performing worse than random guessing. This may suggest a systematic error or a prediction logic error, as it consistently assigns high probabilities to a wrong class.
- **Model D:** This is the better model in the set. Achieves a high F1-Score (0.85) by balancing high Precision (0.83) and high Recall (0.88).

Regression Problems

Evaluation of five regression models (0-4) based on the distance between ground truth (gt) and predicted continuous values.

Calculated Metrics

Metric	Model 0	Model 1	Model 2	Model 3	Model 4
MAE	0.3904	3.9855	12.1268	10.0000	6.3577
MSE	0.2442	24.7629	232.7433	100.0000	281.8465
RMSE	0.4941	4.9762	15.2559	10.0000	16.7883

MedAE	0.3310	3.2562	10.5261	10.0000	3.2355

Analysis of Sensitivity and Outliers

- **Outlier Sensitivity (Model 4 vs. Others):** The Model 4 has a strong sensitivity to outliers. Because its RMSE is the highest out of all the models (16.78), but its MedAE is quite low (3.23). Conveying that although the model is accurate for the majority of data points, it has a few huge errors that are harshly penalized by the quadratic factor of MSE and RMSE.
- **Constant Error (Model 3):** If we take a close look, Model 3's MAE, RMSE, and MedAE (10.0) values are all the same. This is only possible if the absolute error for each prediction is exactly 10.0, meaning a persistent bias/offset in the model.
- **Average vs Median Errors:** The MAE in Models 1 and 2 is greater than their MedAE, this can indicate a distribution of errors that is tilted to the right.

Conclusions

This analysis confirms that no single metric provides a complete picture of the model's performance. In classification, accuracy is misleading if there’s class imbalance (as I stated for Model B), and other cases where F1-Score and AUC-ROC are more reliable. For regression, the choice between MAE and RMSE depends on the tolerance for outliers; RMSE is preferred when large errors are quite undesirable, on the other hand, MedAE provides a better error profile when the data is scattered.

References

1. Code snippets
2. Calculated Metrics - Lab Session Results

classification.py Reg_0.csv

```python
def classificationMetrics(fileName, modelName):  1 usage
    # manage division by 0 errors
    recall = TP / (TP + FN) if (TP + FN) > 0 else 0
    precision = TP / (TP + FP) if (TP + FP) > 0 else 0
    TNR = TN / (TN + FP) if (TN + FP) > 0 else 0
    F1 = 2 * (precision * recall) / (precision + recall) if (precision + recall) > 0 else 0

    dataFrame['rank'] = dataFrame['pred'].rank(method='average')
    R1 = dataFrame[dataFrame['gt'] == 1]['rank'].sum()
    n1 = (dataFrame['gt'] == 1).sum()
    n0 = (dataFrame['gt'] == 0).sum()

    AUC_ROC = (R1 - (n1 * (n1 + 1)) / 2) / (n1 * n0) if (n1 * n0) > 0 else 0
```

```python
def regressionMetrics(fileName, modelName):  1 usage
    df = pd.read_csv(fileName)

    real = df['gt']
    pred = df['pred']

    absError = (real - pred).abs()

    # mean of absolute errors
    mae = absError.mean()

    # mean squared error
    mse = ((real - pred) ** 2).mean()

    # root mean squared error
    rmse = np.sqrt(mse)

    # absolute error median
    medae = absError.median()
```

--- Model A Results ---
TP: 381642 | TN: 118354 | FP: 381646 | FN: 118357
Accuracy: 0.5000
Recall: 0.7633
Precision: 0.5000
TNR: 0.2367
F1-Score: 0.6042
AUC-ROC: 0.4996

--- Model B Results ---
TP: 26622 | TN: 888636 | FP: 11364 | FN: 73378
Accuracy: 0.9153
Recall: 0.2662
Precision: 0.7008
TNR: 0.9874
F1-Score: 0.3859
AUC-ROC: 0.6300

--- Model C Results ---
TP: 108424 | TN: 171906 | FP: 328093 | FN: 391576
Accuracy: 0.2803
Recall: 0.2168
Precision: 0.2484
TNR: 0.3438
F1-Score: 0.2315
AUC-ROC: 0.2753

--- Model D Results ---
TP: 704441 | TN: 62969 | FP: 137029 | FN: 95559
Accuracy: 0.7674
Recall: 0.8806
Precision: 0.8372
TNR: 0.3148
F1-Score: 0.8583
AUC-ROC: 0.6008

--- Model 0 Regression Results ---
MAE: 0.3904
MSE: 0.2442
RMSE: 0.4941
MedAE: 0.3310

--- Model 1 Regression Results ---
MAE: 3.9855
MSE: 24.7629
RMSE: 4.9762
MedAE: 3.2562

--- Model 2 Regression Results ---
MAE: 12.1268
MSE: 232.7433

RMSE: 15.2559
MedAE: 10.5261

--- Model 3 Regression Results ---
MAE: 10.0000
MSE: 100.0000
RMSE: 10.0000
MedAE: 10.0000

--- Model 4 Regression Results ---
MAE: 6.3577
MSE: 281.8465
RMSE: 16.7883
MedAE: 3.2355